POWER PULSE

Corona Important Formulae

Notes by Balaji Sir

1 Critical voltages

1. $$g_0 = 30\ \text{kV/cm (peak)} = 21.1\ \text{kV/cm (rms)}$$

breakdown strength of air at 76 cm Hg, 25 °C

2. $$\delta = \frac{3.92\,b}{273 + t}$$

air density factor, b = pressure in cm Hg, t in °C

3. $$V_c = g_0\,\delta\,m\,r\ln\frac{D}{r} = 21.1\,m\,\delta\,r\ln\frac{D}{r}\ \text{kV (rms, phase)}$$

m = surface irregularity factor (0.8–1)

4. $$V_v = 21.1\,m_v\,\delta\,r\left(1 + \frac{0.3}{\sqrt{\delta r}}\right)\ln\frac{D}{r}$$

visual critical voltage (r in cm)

5. $$g_{max} = \frac{V}{r\ln(D/r)}$$

surface gradient; corona starts when g_max ≥ $g_0\delta$

2 Corona loss (Peek)

6. $$P_c = \frac{241}{\delta}(f + 25)\sqrt{\frac{r}{D}}\,(V - V_c)^2 \times 10^{-5}\ \ \text{kW/km/phase}$$

V and V_c: rms phase voltages in kV; valid for V > V_c

7. $$\text{Foul weather: } V_c \downarrow\ (\approx 0.8V_c)\ \Rightarrow\ \text{loss many times larger}$$

3 Effects and remedies

8. Effects: power loss, radio interference, ozone and nitrous oxide, noise, harmonics

9. $$\text{Reduced by: larger } r,\ \text{larger } D,\ \text{bundled / hollow conductors}$$

10. $$V_c \propto r\ln(D/r)\ \ \Rightarrow\ \ \text{smaller spacing increases corona}$$